**Q1. JEE Main 2026 (24 January Shift 2)**

One mole of $Cl_2(g)$ was passed into 2 L of cold 2 M KOH solution. After the reaction, the concentrations of $Cl^-$, $ClO^-$ and $OH^-$ are respectively (assume volume remains constant)

(1) $0.5M, 0.5M, 1M$
(2) $0.75M, 0.75M, 1M$
(3) $0.5M, 0.5M, 0.5M$
(4) $1M, 1M, 1M$

**Q2. JEE Main 2026 (22 January Shift 2)**

Identify the correct statements :

A. Hydrated salts can be used as primary standard.

B. Primary standard should not undergo any reaction with air.

C. Reactions of primary standard with another substance should be instantaneous and stoichiometric.

D. Primary standard should not be soluble in water.

E. Primary standard should have low relative molar mass.

Choose the correct answer from the options given below :

(1) A, B, C and E Only
(2) A, B and E Only
(3) A, B and C Only
(4) D and E Only

**Q3. JEE Main 2026 (23 January Shift 2)**

200 cc of $x \times 10^{-3}$M potassium dichromate is required to oxidise 750 cc of 0.6 M Mohr's salt solution in acidic medium. Here $x =$ ____.

**Q4. JEE Main 2026 (28 January Shift 1)**

500 mL of 1.2 M KI solution is mixed with 500 mL of $0.2MKMnO_4$ solution in basic medium. The liberated iodine was titrated with standard $0.1MNa_2S_2O_3$ solution in the presence of starch indicator till the blue color disappeared. The volume (in L) of $Na_2S_2O_3$ consumed is ____. (Nearest integer)

**Q5. JEE Main 2026 (24 January Shift 1)**

X and Y are the number of electrons involved, respectively during the oxidation of $I^-$ to $I_2$ and $S^{2-}$ to S by acidified $K_2Cr_2O_7$. The value of $X + Y$ is ____.

## ANSWERS AND SOLUTIONS

**1.** (1) **2.** (3) **3.** 375 **4.** 3 **5.** 12

**1.** **(1)** $Cl_2$ undergoes disproportionation in cold KOH: $Cl_2 + 2OH^- \rightarrow Cl^- + ClO^- + H_2O$
Initial: 1 mole $Cl_2$, 2 L × 2 M KOH = 4 moles $OH^-$
Molar ratio $Cl_2$:$OH^-$ = 1:2 (stoichiometric)
Products formed: 1 mole $Cl^-$, 1 mole $ClO^-$; Excess $OH^-$ = 4 - 2 = 2 moles
In 2 L solution: [$Cl^-$] = 0.5 M, [$ClO^-$] = 0.5 M, [$OH^-$] = 1 M

**2.** **(3)** Primary standard must be soluble for standard solution formation

**3.** **(375)** Mohr's salt ($FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$) is a source of $Fe^{2+}$ ions.
The redox reaction with potassium dichromate in acidic medium is:
$6Fe^{2+} + Cr_2O_7^{2-} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$.
Moles of $Fe^{2+}$ from Mohr's salt solution:
$n(Fe^{2+}) = 0.6 \text{ M} \times 0.750 \text{ L} = 0.45 \text{ mol}$.
From the stoichiometry, 6 moles of $Fe^{2+}$ require 1 mole of $Cr_2O_7^{2-}$.
Moles of $Cr_2O_7^{2-}$ required:
$n(Cr_2O_7^{2-}) = \frac{0.45}{6} = 0.075 \text{ mol}$.
Molarity of potassium dichromate:
$M = \frac{0.075 \text{ mol}}{0.200 \text{ L}} = 0.375 \text{ M} = 375 \times 10^{-3} \text{ M}$.
Therefore, $x = 375$.

**4.** **(3)** $MnO_4^- + I^- \rightarrow MnO_2 + I_2$
$I_2 + S_2O_3^{2-} \rightarrow S_4O_6^{2-} + I^-$
gram eq of $KMnO_4$ = gram eq of $Na_2S_2O_3$
$0.2 \times \dfrac{500}{1000} \times 3 = 0.1 \times V \times 1$
$V = 3 \text{ L}$

**5.** **(12)** In acidic medium, $K_2Cr_2O_7$ acts as a strong oxidizing agent where $Cr_2O_7^{2-}$ is reduced to $Cr^{3+}$.
The reduction half-reaction is: $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$.
For the oxidation of $I^-$ to $I_2$:
The balanced oxidation half-reaction is $2I^- \rightarrow I_2 + 2e^-$.
To balance the electrons with the reduction half-reaction (6 electrons), we multiply the oxidation half-reaction by 3:
$6I^- \rightarrow 3I_2 + 6e^-$.
Thus, the number of electrons involved in the balanced redox reaction for $I^-$ oxidation is $X = 6$.
For the oxidation of $S^{2-}$ to $S$:
The balanced oxidation half-reaction is $S^{2-} \rightarrow S + 2e^-$.

To balance the electrons with the reduction half-reaction (6 electrons), we multiply the oxidation half-reaction by 3:

$3S^{2-} \rightarrow 3S + 6e^-$.

Thus, the number of electrons involved in the balanced redox reaction for $S^{2-}$ oxidation is $Y = 6$.

The value of $X + Y = 6 + 6 = 12$.